Greeks by Monte Carlo: Bumping, Pathwise, and Likelihood Ratio

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Abstract

We compute the European-call Greeks (Delta, Vega, Gamma) by Monte Carlo using three increasingly sophisticated techniques: bumping (finite differences applied to the pricer), pathwise sensitivities (differentiation of the payoff under the integral), and likelihood ratio (differentiation of the kernel of the underlying density). The pathwise and likelihood-ratio estimators are unbiased for the appropriate Greeks under appropriate regularity conditions, while central-difference bumping carries an $O(h^2)$ bias in the bump size h; the three trade off variance, bias, and applicability differently. We derive each from first principles, implement them under the exact GBM sampler, and validate against closed-form Black-Scholes Greeks at the canonical ATM point. The note also surfaces a delicate but important point: the pathwise method does *not* apply to Gamma (the kink of $\max(\cdot, 0)$ produces a Dirac contribution after the second derivative), and the likelihood-ratio method is known to be high-variance for Gamma; in practice Gamma is computed by central finite differences.

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1 The setting

The price of a European call under risk-neutral GBM is

$$C(S_0,\sigma) \;=\; e^{-rT}\,\mathbb{E}\big[\max(S_T-K,0)\big], \qquad S_T \;=\; S_0\exp\big((r-\tfrac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z\big), \qquad Z\sim\mathcal{N}(0,1). \tag{1}$$

The Greeks of interest are

$$\Delta \;=\; \frac{\partial C}{\partial S_0}, \qquad \nu \;=\; \frac{\partial C}{\partial \sigma}, \qquad \Gamma \;=\; \frac{\partial^2 C}{\partial S_0^2}. \tag{2}$$

Black-Scholes provides closed forms (see the note on the Black–Scholes Greeks):

$$\Delta^{\mathrm{BS}} = \Phi(d_1), \qquad \nu^{\mathrm{BS}} = S_0\phi(d_1)\sqrt{T}, \qquad \Gamma^{\mathrm{BS}} = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}, \tag{3}$$

with ϕ the standard normal pdf and $d_1 = (\log(S_0/K) + (r+\sigma^2/2)T)/(\sigma\sqrt{T})$. We use these as ground truth.

The exact GBM MC pricer produces an estimate $\hat{C}_n = \frac{1}{n}\sum_{i=1}^n e^{-rT}\max(S_T^{(i)}-K,0)$ where $S_T^{(i)} = S_0\exp((r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z_i)$ for Z_i iid $\mathcal{N}(0,1)$. Greeks are MC estimators of $\partial C/\partial\theta$ for $\theta\in\{S_0,\sigma\}$ or of the second derivative.

2 Bumping (finite differences)

The conceptually simplest Greek estimator: differentiate the *pricer*, not the payoff.

2.1 Construction

Let $\hat{C}_n(\theta)$ be the MC estimate at parameter θ. Choose a small bump h. The forward-difference estimator of $\partial C/\partial\theta$ is

$$\widehat{\partial_\theta C}_n^{\mathrm{FD}} \;=\; \frac{\hat{C}_n(\theta+h)-\hat{C}_n(\theta)}{h}. \tag{4}$$

The central-difference estimator is

$$\widehat{\partial_\theta C}_n^{\mathrm{CD}} \;=\; \frac{\hat{C}_n(\theta+h)-\hat{C}_n(\theta-h)}{2h}. \tag{5}$$

For Gamma we apply central differences twice:

$$\widehat{\Gamma}_n \;=\; \frac{\hat{C}_n(S_0+h)-2\hat{C}_n(S_0)+\hat{C}_n(S_0-h)}{h^2}. \tag{6}$$

2.2 Common random numbers (CRN)

Naively, the two evaluations $\hat{C}_n(\theta+h)$ and $\hat{C}_n(\theta-h)$ would use independent Monte Carlo samples. The variance of the difference is then $\mathrm{Var}(\hat{C}_n(\theta+h)) + \mathrm{Var}(\hat{C}_n(\theta-h)) \approx 2\sigma^2/n$, and dividing by $2h$ amplifies this further. The result: a bump-and-revalue Greek has variance scaling as $\sigma^2/(nh^2)$, which blows up as $h\to 0$.

The standard fix is *common random numbers*: use the same set of $\{Z_i\}$ for both evaluations. Pathwise on each i, the only thing that changes between the two evaluations is the parameter, not the noise. The variance of the difference is then much smaller because the noise mostly cancels. For our setting,

$$\mathrm{Var}\Big(\hat{C}_n(S_0+h,\{Z_i\}) - \hat{C}_n(S_0-h,\{Z_i\})\Big) \;\approx\; \frac{4h^2}{n}\cdot\mathrm{Var}(\partial_{S_0} f_i) \tag{7}$$

to leading order in h, where $f_i = e^{-rT}\max(S_T^{(i)} - K, 0)$ is viewed as a function of S_0 with Z_i fixed: the n paths are independent, and each per-path difference equals $2h\,\partial_{S_0} f_i$ except on paths for which $S_T^{(i)}$ crosses K between the bumped spots, an event of probability $O(h)$. Dividing by $(2h)^2$, the central-difference Delta estimator has variance $\mathrm{Var}(\partial_{S_0} f_i)/n$: the factor h^2 cancels, and the per-path variance is $O(1)$ in h, that of the pathwise Delta of Section 3. The cancellation does not carry over to Gamma, whose per-path variance is $O(1/h)$ (see below).

2.3 The bias-variance trade-off in h

With CRN, the bumping estimator of Delta or Vega has:

- Bias: $O(h)$ for forward differences, $O(h^2)$ for central differences.
- Variance: roughly constant in h (after CRN cancellation).

The optimal h minimises bias^2 + variance. For central differences the mean squared error is $c\,h^4 + \sigma_\Delta^2/n$, where the per-path variance σ_Δ^2 does not depend on h to leading order, so the error decreases as $h \to 0$ and the only lower limit on h is floating-point round-off. The round-off-optimal step for a central first difference is of order $\epsilon^{1/3}$, where ϵ is the floating-point precision ($\sim 10^{-16}$), so $h_\star \approx 5 \times 10^{-6}$ (in relative units) for double precision. We use $h = 10^{-2}$; the bias is then $\sim 10^{-4}$, comfortably below the half-width at our budgets.

For Gamma, the second-order central difference has bias $O(h^2)$ but per-path variance $O(1/h)$ even with CRN [3, Section 7.1.2]. For the call, the per-path second difference $f_i(S_0+h) - 2f_i(S_0) + f_i(S_0-h)$ vanishes unless $S_T^{(i)}$ crosses K between the bumped spots, an event of probability $O(h)$, and is then of size $O(h)$; after division by h^2 its second moment is $O(h) \cdot O(h^2)/h^4 = O(1/h)$. Gamma is intrinsically harder.

3 Pathwise sensitivities

The bumping method differentiates the pricer. The pathwise method differentiates the *payoff*.

3.1 The interchange of differentiation and expectation

Suppose $f(\theta, Z)$ is the payoff (a function of the parameter θ and the underlying randomness Z), and we want to estimate $\partial_\theta \mathbb{E}[f(\theta, Z)]$. If f is sufficiently regular, we can interchange differentiation and expectation:

$$\frac{\partial}{\partial\theta}\mathbb{E}[f(\theta, Z)] \;=\; \mathbb{E}\left[\frac{\partial f(\theta, Z)}{\partial\theta}\right]. \tag{8}$$

The Monte Carlo estimator of the right-hand side is

$$\widehat{\partial_\theta C}_n^{\mathrm{PW}} \;=\; \frac{1}{n}\sum_{i=1}^{n} \frac{\partial f(\theta, Z_i)}{\partial\theta}. \tag{9}$$

This is unbiased for the true derivative if the interchange is justified, which by dominated convergence requires f Lipschitz in θ uniformly in Z (or weaker conditions). The European call payoff $\max(S_T - K, 0)$ is Lipschitz in S_T (with constant 1), which is in turn C^∞ in S_0 and σ, so the interchange is valid.

3.2 Pathwise Delta and Vega for the European call

Compute $\partial f/\partial S_0$ first. Using

$$S_T = S_0 \exp\big((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\big) \;=\; \frac{S_T}{S_0}\cdot S_0, \tag{10}$$

so $\partial S_T/\partial S_0 = S_T/S_0$. By the chain rule (and noting that $\max(x,0)$ has derivative $\mathbf{1}_{\{x>0\}}$ almost everywhere),

$$\frac{\partial f}{\partial S_0} \;=\; e^{-rT}\cdot \mathbf{1}_{\{S_T>K\}}\cdot \frac{S_T}{S_0}. \tag{11}$$

For Vega:

$$\frac{\partial S_T}{\partial \sigma} \;=\; S_T\cdot(-\sigma T + \sqrt{T}\,Z), \tag{12}$$

so

$$\frac{\partial f}{\partial \sigma} \;=\; e^{-rT}\cdot \mathbf{1}_{\{S_T>K\}}\cdot S_T\cdot(\sqrt{T}\,Z - \sigma T). \tag{13}$$

The pathwise Delta and Vega estimators are then

$$\widehat{\Delta}_n^{\mathrm{PW}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\mathbf{1}_{\{S_T^{(i)}>K\}}\frac{S_T^{(i)}}{S_0}, \tag{14}$$

$$\widehat{\nu}_n^{\mathrm{PW}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\mathbf{1}_{\{S_T^{(i)}>K\}}S_T^{(i)}(\sqrt{T}\,Z_i - \sigma T). \tag{15}$$

These are unbiased and have variance comparable to (often less than) the variance of the price estimator itself. They are the gold standard when applicable.

3.3 Why pathwise fails for Gamma

For Gamma we want $\partial^2 C/\partial S_0^2$, which would require differentiating $\partial f/\partial S_0$ once more. Computing:

$$\frac{\partial^2 f}{\partial S_0^2} \;=\; \frac{\partial}{\partial S_0}\big[e^{-rT}\mathbf{1}_{\{S_T>K\}}\frac{S_T}{S_0}\big]. \tag{16}$$

The indicator $\mathbf{1}_{\{S_T>K\}}$ is the Heaviside step function of $S_T - K$. Its derivative in the distributional sense is the Dirac delta $\delta(S_T - K)$. Once we differentiate, the expectation

$$\mathbb{E}[\delta(S_T - K)\cdot(\text{stuff})] \tag{17}$$

is finite (it equals (stuff at $S_T = K$) times the density of S_T at K), but *not estimable by Monte Carlo*: a single sample has $S_T = K$ with probability zero, so the sum is identically zero on every realisation.

This is the canonical failure of pathwise for second-order Greeks of payoffs with kinks: the second derivative is a measure, not a function, and Monte Carlo can only estimate functions.

4 Likelihood ratio

The likelihood-ratio (LR) method, which predates its use for option Greeks (Glynn, 1987; Reiman and Weiss, 1989) and was applied to them by Broadie and Glasserman [1], sidesteps the problem of differentiating the payoff by differentiating the *density of the underlying* instead.

4.1 The construction

Suppose S_T has a density $p(s;\theta)$ depending on the parameter θ. Then

$$C(\theta) \;=\; e^{-rT}\int f(s)\,p(s;\theta)\;\mathrm{d}s, \tag{18}$$

and (assuming we can interchange differentiation and integration)

$$\frac{\partial C}{\partial\theta} \;=\; e^{-rT}\int f(s)\cdot\frac{\partial p(s;\theta)/\partial\theta}{p(s;\theta)}\cdot p(s;\theta)\;\mathrm{d}s \;=\; e^{-rT}\mathbb{E}[f(S_T)\cdot\mathcal{S}(\theta,S_T)]\,, \tag{19}$$

where

$$\mathcal{S}(\theta,S_T) \;\overset{\text{def}}{=}\; \frac{\partial}{\partial\theta}\log p(S_T;\theta) \tag{20}$$

is the *score function* associated with the density family.

The MC estimator is

$$\widehat{\partial_\theta C}_n^{\mathrm{LR}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\,f(S_T^{(i)})\cdot\mathcal{S}(\theta,S_T^{(i)}). \tag{21}$$

4.2 Score function for GBM

Under GBM, $\log S_T = \log S_0 + (r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z$ is normal with mean $\mu = \log S_0 + (r-\frac{1}{2}\sigma^2)T$ and standard deviation $\nu = \sigma\sqrt{T}$. The density of $\log S_T$ is

$$p_{\log}(\ell;S_0,\sigma) \;=\; \frac{1}{\nu\sqrt{2\pi}}\exp\!\left(-\frac{(\ell-\mu)^2}{2\nu^2}\right). \tag{22}$$

Differentiating $\log p_{\log}$ with respect to S_0:

$$\mathcal{S}_{S_0}(\log S_T) \;=\; \frac{Z}{S_0\sigma\sqrt{T}}. \tag{23}$$

With respect to σ:

$$\mathcal{S}_{\sigma}(\log S_T) \;=\; \frac{Z^2-1}{\sigma} - Z\sqrt{T}. \tag{24}$$

(Both derivations use the chain rule and the fact that $Z = (\log S_T - \mu)/\nu$.)

The LR Delta and Vega estimators are

$$\widehat{\Delta}_n^{\mathrm{LR}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\max(S_T^{(i)}-K,0)\cdot\frac{Z_i}{S_0\sigma\sqrt{T}}, \tag{25}$$

$$\widehat{\nu}_n^{\mathrm{LR}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\max(S_T^{(i)}-K,0)\cdot\left(\frac{Z_i^2-1}{\sigma} - Z_i\sqrt{T}\right). \tag{26}$$

4.3 Variance comparison: pathwise vs LR

For the European call, both pathwise and LR are unbiased. The variance comparison is:

- Pathwise: the estimand is $e^{-rT}\mathbf{1}_{\{S_T>K\}}(S_T/S_0)$, which is bounded for typical paths and has moderate variance.
- LR: the estimand is $f(S_T)\cdot Z/(S_0\sigma\sqrt{T})$, which carries the full payoff f multiplied by a factor that can have large absolute value when $|Z|$ is large. Tails of Z contribute $f(S_T)\cdot Z$, which can be large.

Empirically, for the call at the canonical ATM point, the per-path variance of the pathwise Delta is about 6.4 times smaller than that of the LR Delta (0.332 against 2.135, Test 2 of `validate_greeks_mc.py`), and about 12.9 times smaller for Vega (Test 3); the corresponding half-width ratios are 2.5 and 3.6. The advantage of LR is universality: for digital options where f is a Heaviside (not Lipschitz), pathwise fails but LR works.

5 Algorithmic specifications

Algorithm 1 Bump-and-revalue Delta with central differences and CRN

Require: Spot S_0, parameters, sample size n, bump h, rng.

Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.

$C^+ \leftarrow \frac{1}{n}\sum_i e^{-rT}\max(S_0(1+h)e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z_i} - K, 0)$

$C^- \leftarrow \frac{1}{n}\sum_i e^{-rT}\max(S_0(1-h)e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z_i} - K, 0)$

$\Delta \leftarrow (C^+ - C^-)/(2S_0h)$

return Δ

The trick: we bump S_0 multiplicatively (by $1 \pm h$) so that the formulae work uniformly without recomputing the diffusion coefficient.

Algorithm 2 Pathwise Delta

Require: Spot S_0, parameters, sample size n, rng.

Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.

for $i = 1, \dots, n$ **do**

  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i)$

  $\Delta_i \leftarrow e^{-rT} \cdot \mathbf{1}_{\{S_T^{(i)} > K\}} \cdot S_T^{(i)}/S_0$

end for

return $\bar{\Delta} = n^{-1}\sum_i \Delta_i$

Algorithm 3 Likelihood-ratio Delta

Require: Spot S_0, parameters, sample size n, rng.

Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.

for $i = 1, \dots, n$ **do**

  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i)$

  $\Delta_i \leftarrow e^{-rT}\max(S_T^{(i)} - K, 0) \cdot Z_i/(S_0\sigma\sqrt{T})$

end for

return $\bar{\Delta} = n^{-1}\sum_i \Delta_i$

6 Empirical validation strategy

The script `python/validate_greeks_mc.py` runs four tests.

Test 1: BS coherence. For each Greek and each method, the estimator at $n = 10^5$ should agree with the closed-form BS Greek within a few standard errors.

Test 2: variance comparison Delta. Pathwise vs LR vs bumping. Predicted: bumping $\approx$ pathwise $<$ LR; measured per-path variances 0.326 (bumping), 0.332 (pathwise) and 2.135 (LR). With CRN, the central difference converges path by path to the pathwise derivative as $h \to 0$, so at small h the two have nearly the same variance. Report half-width ratios at fixed budget.

Test 3: variance comparison Vega. Same as Test 2; measured half-widths 0.4685 (bumping), 0.4686 (pathwise) and 1.684 (LR).

Test 4: input validation. Mechanical: bad inputs should raise.

The implementation is a single module `quantlib.greeks` containing seven estimators: bumping, pathwise and LR for Delta and Vega (`delta_bump`, `delta_pathwise`, `delta_lr` and the corresponding `vega_` functions), and bumping only for Gamma (`gamma_bump`); there is no pathwise or LR Gamma.

7 Bibliographic notes

The pathwise method for option Greeks is due to Broadie and Glasserman [1], who also apply the likelihood-ratio method to option Greeks; the broader context of efficient simulation of derivatives is in [3, Chapter 7]. The malliavin calculus generalisation, which provides a unified framework for both pathwise and LR, is in Fournié et al. [2]. We do not pursue the Malliavin generalisation in this writeup but note that it is the modern theoretical lens.

References

[1] M. Broadie and P. Glasserman, *Estimating security price derivatives using simulation*, Management Science, 42(2):269–285, 1996.

[2] E. Fournié, J. M. Lasry, J. Lebuchoux, P. L. Lions, N. Touzi, *Applications of Malliavin calculus to Monte-Carlo methods in finance*, Finance and Stochastics, 3(4):391–412, 1999.

[3] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.